VANISHA SHARMA

M.Sc. Biotechnology | Dissertation Trainee at CSIR-CDRI
Mail | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

MSc Biotechnology graduate with strong research training in endometriosis, reproductive inflammation, and molecular and cellular biology using in vitro and in vivo models. Experienced in investigating immune and signalling pathways relevant to fertility, disease progression, and pregnancy outcomes.

EDUCATION

Manipal University, Jaipur
Master of Science
Biotechnology, 2023-2025
CGPA: 7.91/10

The IIS (deemed to be) University
Bachelor's in Science and Education
Life Science and Education, 2019-2023
CGPA: 8.29/10

TECHNICAL SKILLS

Molecular & Cell Biology: Mammalian cell culture (ovarian & endometrial cells), RNA isolation, cDNA synthesis, RT-PCR, SDS-PAGE, Western blotting, plasmid purification, fluorescence microscopy
Model Systems: Surgical endometriosis mouse model and DHEA-induced PCOS mouse model; ethical animal handling, dosing, and tissue collection
Functional & Mechanistic Studies: Overexpression systems (RhoG/RhoH), inflammatory chemokine analysis, compound perturbation (Danazol, UPF1069, chloroquine, ILK modulation)
Computational / Bioinformatics: R for statistics and visualization, image-based quantification, omics data analysis training
Software & Tools: GraphPad Prism, ImageJ, Adobe Illustrator, BioRender, Cytoscape

RESEARCH EXPERIENCE

Computational & Bioinformatics Research Experience
AI and Bioinformatics (Online Module) — **August, 2025- December, 2025**
Independent / Collaborative Bioinformatics Projects
Project: *Integrated Transcriptomic & Network Analysis of Endometriosis* LINK

- Protein-protein interaction and network analysis of endometriosis-associated genes
- Identification of **epithelial junction and EMT-related modules** using STRING, Cystoscope, and MCODE
- Hub and bridge gene analysis (*PAX2, EPCAM, CLDN7, ESR1*) linked to tissue remodeling and invasiveness.

Project*: Integrated Transcriptomic Analysis of PCOS reveals Metabolic and Endocrine-Transcriptional Axes* LINK

- Integrated **multi-dataset transcriptomic analysis** using microarray (GSE137684, GSE114419, GSE98421) and RNA-seq (GSE277906) datasets.
- Differential expression and co-expression analysis using **limma, DESeq2, and WGCNA.**
- **Cortisol-responsive gene integration** to examine stress-axis involvement in reproductive dysfunction.
- Functional enrichment (GO, KEGG, Reactome) highlighting inflammatory and metabolic pathway dysregulation.

Tools and Methods:
R (limma, DESeq2, WGCNA), GEO, Cystoscope, GO/KEGG, PCA, network analysis

Dissertation Intern, Skill Development Programme — **Jan 2025 – June 2025**

CSIR-CDRI, Lucknow

Project: "Functional Analysis of RhoG/RhoH Overexpression and ILK–ILKAP Axis in Endometriotic Cells"

1. Developed an in vitro overexpression system in IHECs to study RhoG/RhoH functions in cell movement and inflammation pathways
2. Performed cloning, RNA isolation, gene expression profiling, and Western blotting to evaluate ILK/ILKAP signaling changes
3. Established experimental workflows with scope for future validation and translational research
4. Linked chemical modulation with endometrial cell signaling responses

Summary Training, Skill Development Programme **May 2024 – Sept 2024**

CSIR CDRI, Lucknow

Project: "Role of PARP-2 in Regulating Inflammatory Chemokines in Endometriotic Cells"

1. Investigated PARP-2 inhibition (UPF1069) in IHECs using RNA extraction, cDNA synthesis, qPCR, and Western blotting
2. Engineered 3D spheroid and co-culture models that revealed differential chemokine responses to PARP-2 inhibition and Danazol treatment
3. Conducted compound exposure assays (chloroquine, ILK modulation) to study downstream inflammatory signaling
4. Contributed to cell culture maintenance, troubleshooting, and internal presentations

Hands-on Training on Fundamentals of Animal Handling **September 2024**

CSIR CDRI, Lucknow

1. Trained in ethical handling, dosing, and termination in rodents
2. Assisted in developing DHEA-induced PCOS and surgical endometriosis models
3. Performed ovarian tissue collection for histological and molecular analysis.

PUBLICATIONS

Gupta, S., Tripathi, R., Kawale, A. K., Sarkar, S., Singh, A., Verma, R. K., Sankhwar, P. L., **Sharma, V**., & Jha, R. K. (2025). *PARP-2 acts on ILK signalling, and pharmacological targeting of PARP-2 ameliorate endometriosis in a mouse model*. Biochemical and biophysical research communications, 754, 151509. https://doi.org/10.1016/j.bbrc.2025.151509

CONFERENCES AND WORKSHOP

- **NaviClar–EURAXESS Symposium (2025):** Research presentation and international scientific engagement
- **Next-Generation Sequencing & Precision Oncology Training (2025)**

LEADERSHIP AND OTHER ACTIVITIES

1. Coordinated as Core Committee Member, National Science Day, Manipal University, Jaipur (Feb 2024)
2. Supported as Stage Committee Member, RAMSACT Conference, Manipal University Jaipur (April 2024)
3. Edited departmental newsletter as Student Editor, Bioscience Patrika (Jun 2024–Dec 2024)
4. Anchored the 74th CSIR-CDRI Annual Day (Feb 2025)

REFERENCES

1. **Dr Sandeep K. Srivastava**
   Associate Professor and Head at Manipal University, Jaipur
   Email: sandeepkumar.srivastava@jaipur.manipal.edu
2. **Dr. Mousumi Debnath**
   Associate Professor at Manipal University, Jaipur
   Email: mousumi.debnath@jaipur.manipal.edu